

Recurrence Relations — Solutions

Instructor solution key.

The Master Theorem (Roughgarden)

Let $a \geq 1$, $b \geq 2$, and $d \geq 0$ be constants, and suppose $T(n)$ is defined on the nonnegative integers by the recurrence

$$T(n) \leq a \cdot T(n/b) + O(n^d), \quad T(1) \leq c,$$

where n/b means either $\lceil n/b \rceil$ or $\lfloor n/b \rfloor$, and c is a constant. Then

$$T(n) = \begin{cases} O(n^d \log n) & \text{if } a = b^d \quad [\text{Case 1}] \\ O(n^d) & \text{if } a < b^d \quad [\text{Case 2}] \\ O(n^{\log_b a}) & \text{if } a > b^d \quad [\text{Case 3}] \end{cases}$$

Note: each recurrence below is stated with equality, so once the matching case pins down the exponent, the same argument gives a matching lower bound — so we state the final answer as $\Theta(\cdot)$ rather than just $O(\cdot)$.

Worked Example

Here is a fully worked example showing how to apply the theorem, in the same format used for the solutions below.

Example (Binary Search). Solve $T(n) = T(n/2) + 1$.

Identify a , b , $f(n)$: $a = 1$, $b = 2$, $f(n) = 1 = O(n^0)$, so $d = 0$.

Compare a to b^d : $b^d = 2^0 = 1$. Since $a = 1 = b^d$, we are in **Case 1**.

Apply the theorem: Case 1 gives $T(n) = O(n^d \log n) = O(n^0 \log n) = O(\log n)$.

Final Answer: $T(n) = \Theta(\log n)$ <i>(Case 1: $a = b^d$)</i>

Problem 1. $T(n) = 2T(n/2) + 3n$

Identify a , b , $f(n)$: $a = 2$, $b = 2$, $f(n) = 3n = O(n^1)$, so $d = 1$.

Compare a to b^d : $b^d = 2^1 = 2$. Since $a = 2 = b^d$, we are in **Case 1**.

Apply the theorem: Case 1 gives $T(n) = O(n^d \log n) = O(n \log n)$.

Final Answer: $T(n) = \Theta(n \log n)$

(Case 1: $a = b^d$)

Problem 2. $T(n) = 3T(n/4) + \sqrt{n}$

Identify a , b , $f(n)$: $a = 3$, $b = 4$, $f(n) = \sqrt{n} = O(n^{1/2})$, so $d = \frac{1}{2}$.

Compare a to b^d : $b^d = 4^{1/2} = 2$. Since $a = 3 > 2 = b^d$, we are in **Case 3**.

Apply the theorem: Case 3 gives $T(n) = O(n^{\log_b a}) = O(n^{\log_4 3})$, where $\log_4 3 \approx 0.7925$.

Final Answer: $T(n) = \Theta(n^{\log_4 3})$

(Case 3: $a > b^d$)

Problem 3. $T(n) = 7T(n/2) + \Theta(n^3)$

Identify a , b , $f(n)$: $a = 7$, $b = 2$, $f(n) = \Theta(n^3)$, so $d = 3$.

Compare a to b^d : $b^d = 2^3 = 8$. Since $a = 7 < 8 = b^d$, we are in **Case 2**.

Apply the theorem: Case 2 gives $T(n) = O(n^d) = O(n^3)$.

Final Answer: $T(n) = \Theta(n^3)$

(Case 2: $a < b^d$)

Summary

Problem	a, b, d	b^d	Case	$T(n)$
1	$a = 2, b = 2, d = 1$	2	1 ($a = b^d$)	$\Theta(n \log n)$
2	$a = 3, b = 4, d = \frac{1}{2}$	2	3 ($a > b^d$)	$\Theta(n^{\log_4 3})$
3	$a = 7, b = 2, d = 3$	8	2 ($a < b^d$)	$\Theta(n^3)$